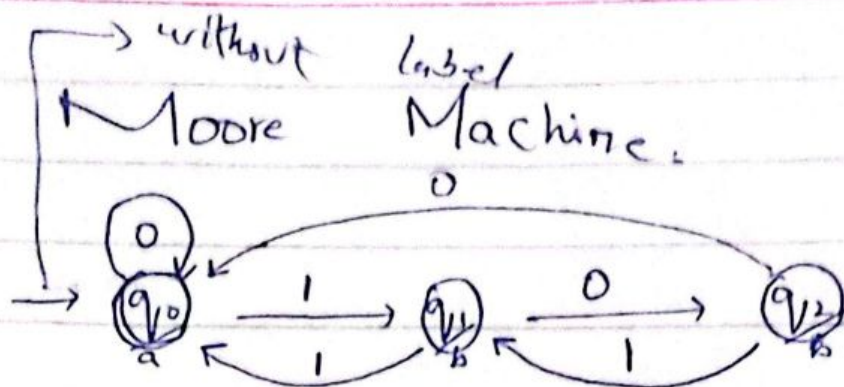


edges = input symbol. $\Sigma = \{0, 1\}$
 blue $\rightarrow$ output symbol.

$$Q = \{q_0, q_1, q_2\}$$

$$\delta \rightarrow Q \times \Sigma(0,1)$$

q_0	$q_0 q_1$
q_1	$q_2 q_0$
q_2	$q_0 q_1$

$$\lambda: Q \rightarrow \Delta$$

q_0	$\rightarrow$	a
q_1	$\rightarrow$	b
q_2	$\rightarrow$	b

phla symbol laemi aygo aygo.

How to process a string & find output against it.

Input-n $\Rightarrow$ 00110 (pass krega dfa me)

$\rightarrow$ a a a b a a $\Rightarrow$ output = n+1

Mam

010101 = n acha n/ke
| | | | |
a a b b b b = n+1

Transition Table.

Current state	Next state		Output
	0	1	
q_0	q_0	q_1	a
q_1	q_2	q_0	b
q_2	q_0	q_1	b

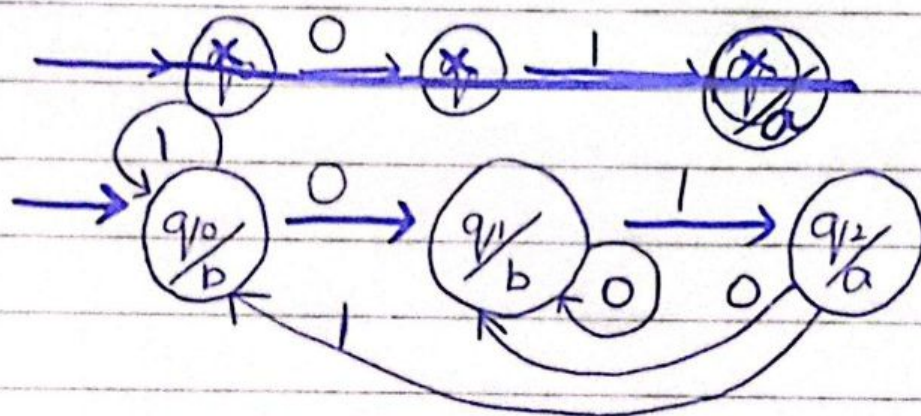
q q_0

How to construct a Moore Machine?

- 1) Construct a MM that prints 'a' whenever sequence '01' is encountered.

$\Sigma = \{a, b\}$, $\Delta = \{0, 1\}$ input

DFA:

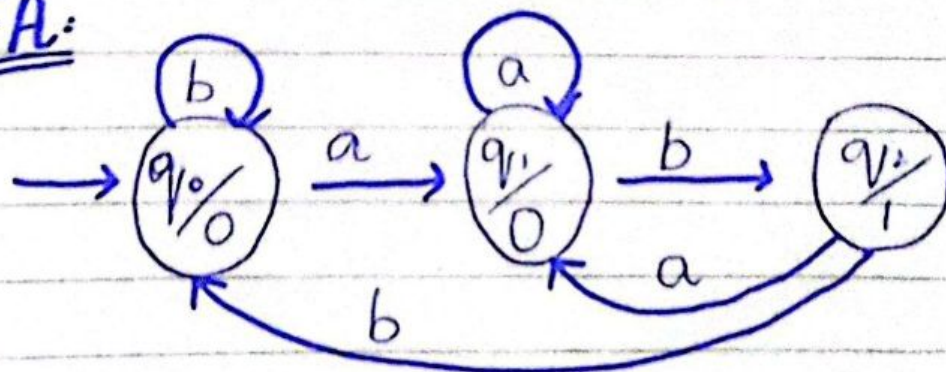


0110 $\Rightarrow$ bba bb (n+1) output

0101 $\Rightarrow$ bba ba

- 2) Construct a MM that prints '1' whenever the sequence 'ab' is encountered in any input string.

DFA:

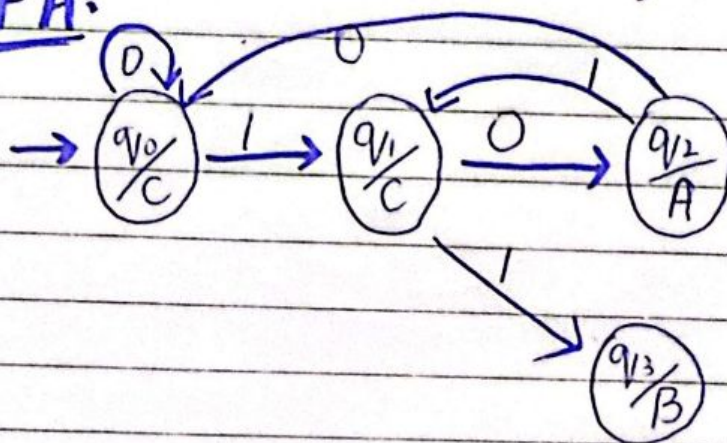


Day: _____

Date: _____

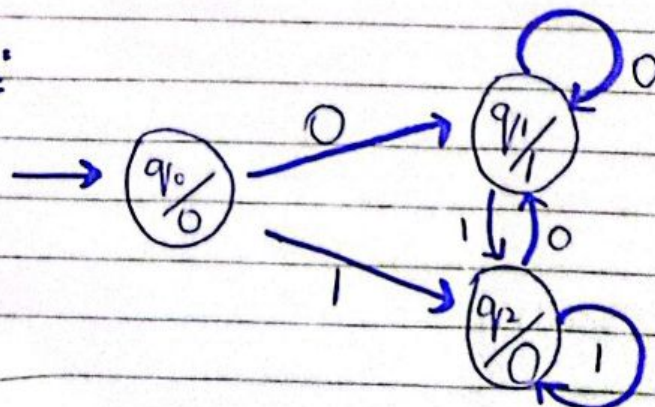
- 3) Design a MM that takes set of all strings over $\{0,1\}$ and produces an 'A' as output IF input ends with '10' or produces 'B' as output if input ends with '11', otherwise process C.
- $\Sigma = \{0,1\}$
 $\Delta = \{A,B,C\}$

DFA.



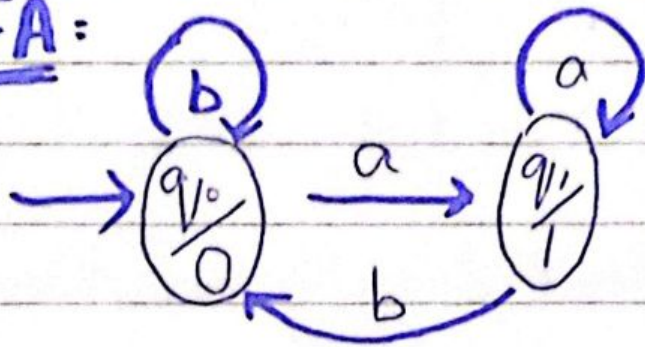
- 4) Design a MM that produces 1's compliment of any binary input string. $\Sigma = \{0,1\}$, $\Delta = \{0,1\}$

DFA.



5) Design a MM that takes all strings of a's & b's as input and counts the no. of a's in the input in terms of $\Sigma = \{a, b\}$, $\Delta = \{0, 1\}$

DFA:



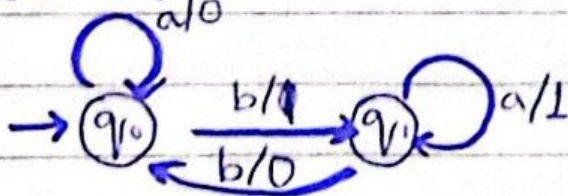
abbabb $\rightarrow$ 0100100

Mealy Machine

Config $\rightarrow$ same as Moore Machine

output function $\rightarrow \lambda: Q \times \Sigma \rightarrow \Delta$

Picture:



$\Sigma = \{a, b\}$

$\Delta = \{0, 1\}$

$q_0 \times a \rightarrow 0$

$q_0 \times b \rightarrow 1$

$q_1 \times a \rightarrow 1$

$q_1 \times b \rightarrow 0$

~~Current S | input = a~~

Day: _____

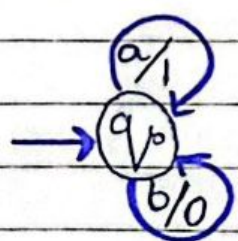
Date: _____

Transition Table:

Current State	Next State			
	input = a		input = b	
	state	output	state	output
q_0	q_0	0	q_1	1
q_1	q_1	1	q_0	0

How to construct Mealy Machine.

- 1) Design a MM that takes all strings of a's & b's and as inputs and counts the no. of a's in the input in terms of 1.

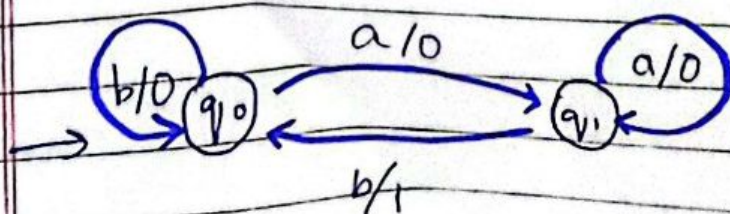


$$\Sigma = \{a, b\}$$

$$\Delta = \{0, 1\}$$

$$\Sigma = ababaab = 1010110 = \Delta$$

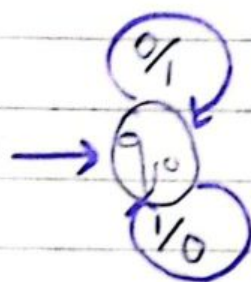
- 2) Construct a MM that prints '1' whenever the sequence 'ab' is encountered.



Day: _____

Date: _____

3) Design a MM that prints 1's complement of every binary input string.

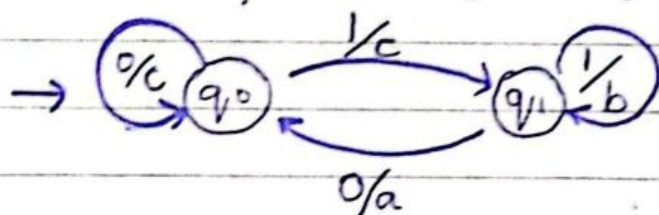


$$\Sigma = \{0, 1\}$$

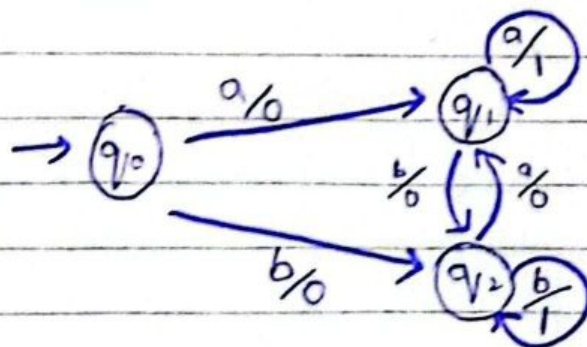
$$\Delta = \{0, 1\}$$

4) $10 \rightarrow a$, $11 \rightarrow b$, else c (00, 01, 10, 11)

$$\Sigma = \{0, 1\}, \Delta = \{a, b, c\}$$



5) Design a MM where $\Sigma = \{a, b\}$ and strings end with either 'aa' or 'bb' giving a output of '1' (ab, ba, a, b) $\rightarrow 0$, aa $\rightarrow 1$, bb $\rightarrow 1$



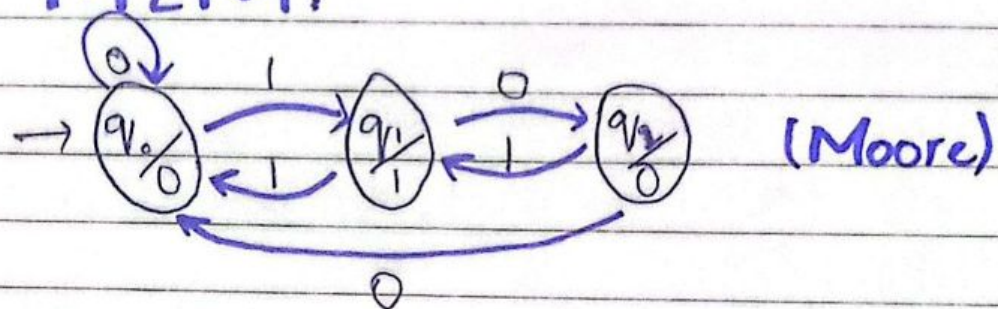
Day: _____

Date: _____

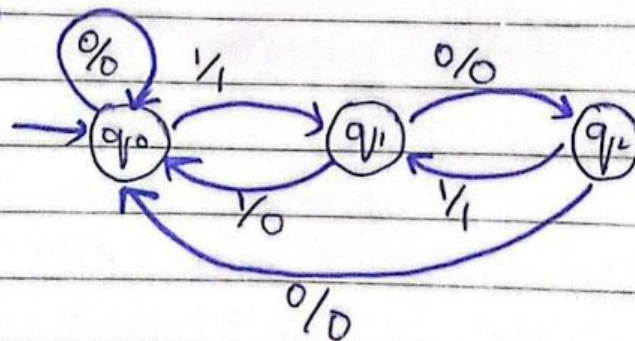
Moore to Mealy Conversion:

- 1- Machine to Machine
- 2- T.T to T.T

1) M2M:



Mealy:



2. T.T → T.T:

Current state	Next state		Output
	0	1	
q_0	q_0	q_1	0
q_1	q_2	q_0	1
q_2	q_0	q_1	0

(Moore)

Day: _____

Date: _____

Mealy:

Current State	Next State			
	input = 0		input = 1	
	state	Output	state	output
q_0	q_0	0	q_1	1
q_1	q_1	0	q_0	0
q_2	q_0	0	q_1	1

Mealy to Moore (pending)

* "Closure Properties" of Regular Lang:

Principle of Closure Properties:

- Union of two R.L is regular
- Complement of R.L is regular
- Intersection of two R.L is regular.
- Closure of a R.L is regular.
- Product / Concatenation of R.L is regular.

Example:

Set of Integers $\Rightarrow \mathbb{Z}$

$$a = 1 \in \mathbb{Z}$$

$$b = 2 \in \mathbb{Z}$$

Addition of a, b

$$a + b = 1 + 2 = 3$$

$(\mathbb{Z}, +)$ = set of integers is closed under Addition. $a, b \in \mathbb{Z}$, $a + b \in \mathbb{Z}$

Day: _____

Date: _____

We can prove these two properties with the help of either "R.E's" or "FA's"

I- Union:

RE:

$$L_1 \rightarrow R.E_1 \rightarrow a^*$$

$$L_2 \rightarrow R.E_2 \rightarrow (a+b)$$

$$\begin{array}{l} L_1 \cup L_2 = R.E_1 + R.E_2 = a^* + (a+b) \\ L_1 + L_2 \quad \quad \quad R.E_3 \rightarrow R.L \end{array}$$

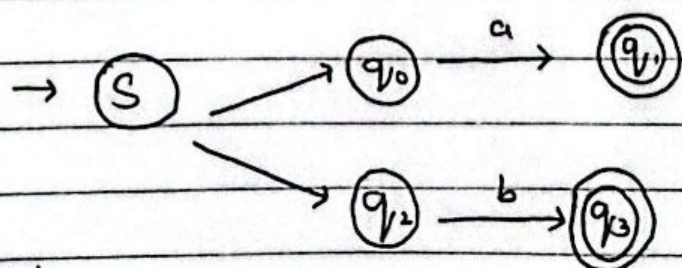
Hence $L_1 + L_2$ is a Regular Lang.
Union is closed under R.L

FA:

$$L_1 \rightarrow \rightarrow (q_0) \xrightarrow{a} (q_1)$$

$$L_2 \rightarrow \rightarrow (q_2) \xrightarrow{b} (q_3)$$

$$\text{Union} = L_1 + L_2 (L_1 \cup L_2)$$



Union is closed under R.L

Day: _____

Date: _____

2- Complement:

$$L_0 \quad \Sigma^*, \Sigma = \{a, b\}$$

$$\overline{L_0} = \Sigma^* - L$$

$$\overline{L}, L^c, L'$$

RE:

$$RE_c = \Sigma^* - R.E$$

FA:

(final to non-final or non-final to final)

$$L \rightarrow \text{DFA} \rightarrow \overline{\text{DFA}}$$

3- Intersection:

$$L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}} \text{ OR } (L_1^c \cup L_2^c)^c \quad \text{😊}$$

4- Closure:

$$L_1 \rightarrow L_1^*$$

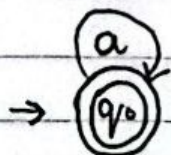
If L is $R.L$ so is L^* .

R.E:

$$L_1 \rightarrow R.E \rightarrow a$$

$$L_1^* \rightarrow a^* \rightarrow RE \rightarrow R.L$$

F.A:



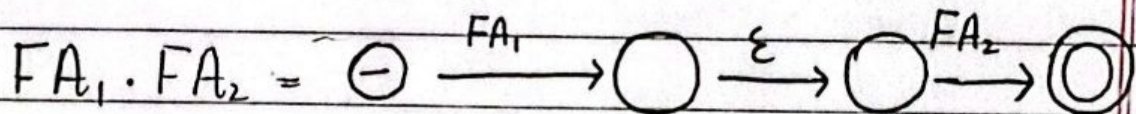
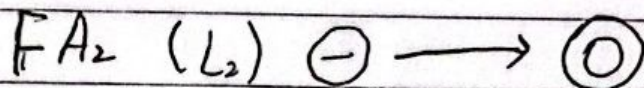
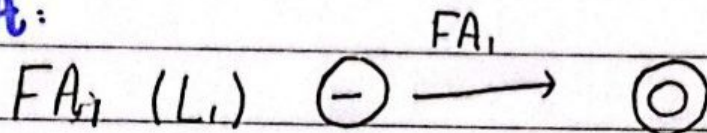
Closed under closure

5- Concatenation:

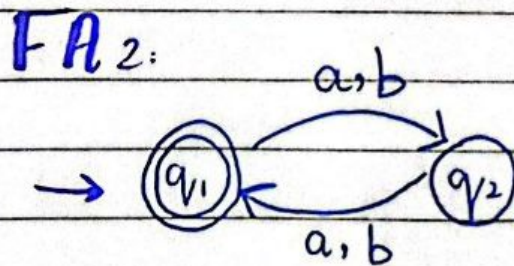
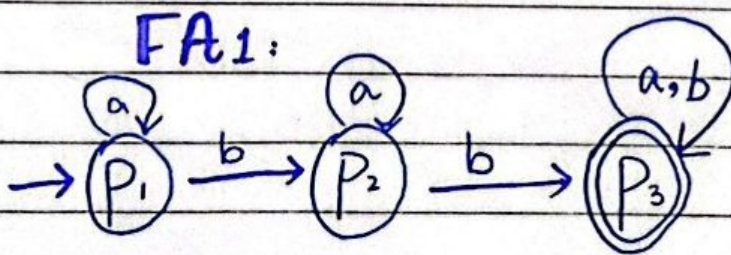
$$L_1 \rightarrow R \cdot E_1, \quad L_2 \rightarrow R \cdot E_2$$

$$L_1 \cdot L_2 = R \cdot E_1 \cdot R E_2 = R \cdot E_3 \Rightarrow R \cdot L$$

F.A:



Intersection Of FA's:-



$FA_1 \cap FA_2$

1)

Day: _____

Date: _____

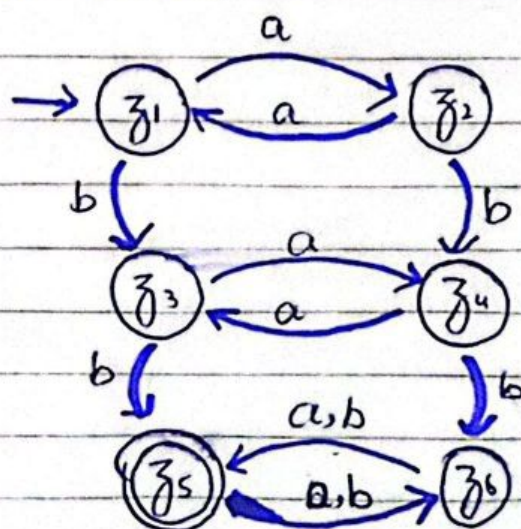
1) Make "combined" Transition Table for FA_1 & FA_2 :-

old state	a	b
$p_1 q_1 = z_1$	$p_1 q_2 = z_2$	$p_2 q_2 = z_3$
$p_1 q_2 = z_2$	$p_1 q_1 = z_1$	$p_2 q_1 = z_4$
$p_2 q_2 = z_3$	$p_2 q_1 = z_4$	$p_3 q_1 = z_5$
$p_2 q_1 = z_4$	$p_2 q_2 = z_3$	$p_3 q_2 = z_6$
$p_3 q_1 = z_5$	$p_3 q_2 = z_6$	$p_3 q_2 = z_6$
$p_3 q_2 = z_6$	$p_3 q_1 = z_5$	$p_3 q_1 = z_5$

Rules for final states:

Mark all those states (as final states) where p_3 & q_1 appear together.

$FA_1 \cap FA_2$:



Day: _____

Date: _____

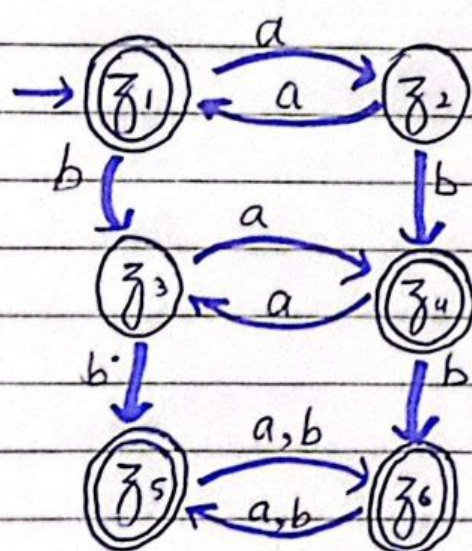
Hence Set of R.L closed under closed under intersection.

Union of FA's:

Rule for final States:

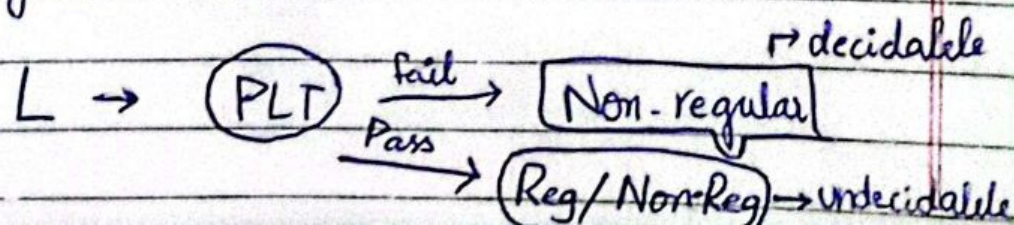
Mark all those states (as final states) where ~~both~~ p_3 OR q_1 appears.

$FA_1 \cup FA_2$



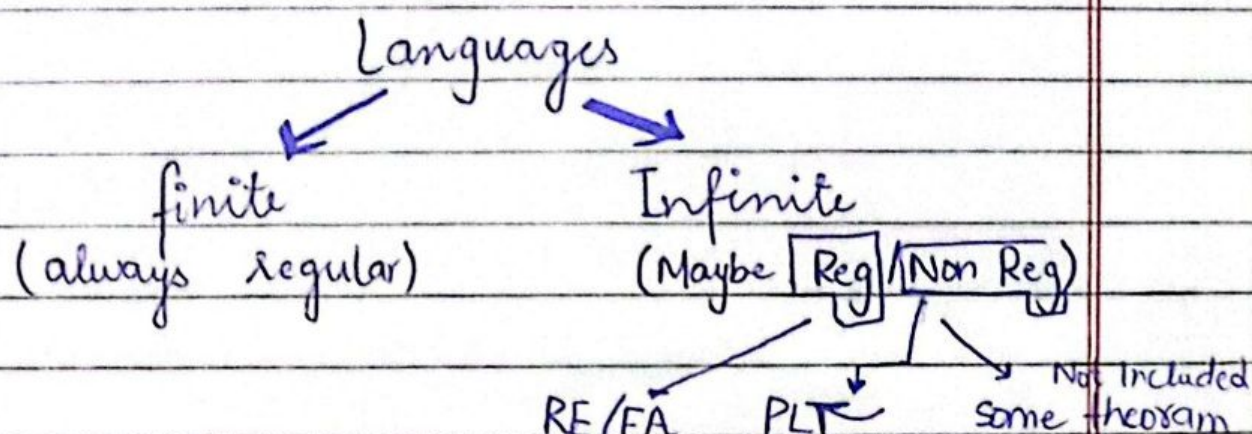
Pumping Lemma:

whether a A test to prove certain lang. are non-regular. It is also called a Negative test.



Day: _____

Date: _____



Q. Prove that lang. L is non-regular.

Proof Steps: (proof by contradiction)

- 1- Suppose that lang. L is regular.
- 2- Then we can make an FA of lang ' L '
↳ Suppose that the FA of ' L ' is N -States.
- 3- Take a string (from ' L ') of length greater than No. of states (N).
 $\text{length}(w) > N$
- 4- Divide the string w into three parts i.e x, y, z such that $\text{len}(x) + \text{len}(y) < N$.
- 5- Also ' x ' & ' z ' can be NULL, but y can't be NULL.
i.e $\text{len}(y) \geq 1$

6- Pump (y) & decide the lang 'L' is non-Reg.

Example:

Sol: Prove $L = a^n b^n$ is non-Reg.

- 1- Assume that 'L' ($a^n b^n$) is Regular.
- 2- An FA for 'L' a FA consist of 8 states ($N=8$)
- 3- Take a string 'w' from 'L' of lang. more than $N=8$.

$$w = \underbrace{aaaaa}_x \underbrace{bb}_y \underbrace{bbbbb}_z$$

- 4- $\text{len}(x)=2$
 $\text{len}(y)=3$
 $\text{len}(x+y) < N$
 $2+3 < 8$
 $5 < 8$

- 5- $y=3$

$$w = xyz$$

6. Pump (y) = $xyyz$

Does " $aaaaaaaaabbbbb$ " $\in L$?

Since " $aaaaaaaaabbbbb$ " does not belong $a^n b^n$, hence L must be Non-Reg.

Day: _____

Date: _____

Example:

Prove Palindrome is Non-Reg.

Sol:

- 1- Assume that $L(\text{Palindrome})$ is Regular.
- 2- An FA for 'L' a FA consist of 7 states ($N=7$)
- 3- Take a string 'w' from L of length more than $N=7$.

$$w = \underbrace{a}_{y} \underbrace{b}_{z} a$$

$$w = \underbrace{a}_{x} \underbrace{b}_{y} \underbrace{a}_{z}$$

4- $\text{len}(x) = 0, \in$
 $\text{len}(y) = 2$
 $\text{len}(x+y) < N$
 $0+2 < 7$
 $2 < 7$

$\text{len}(x) = 1$
 $\text{len}(y) = 1$
 $\text{len}(x+y) < N$
 $2 < 7$

5- $y = 2$

$y = 1$

6- $w = yz$

$w = xyz$

$\text{Pump}(y) = yyz$
 $= ababa$

$\text{Pump}(y) = \underbrace{xyyz}_{\text{ababa}}$
 $= abba$

Day: _____

Date: 22/2/21

Example:

Prove Palindrome is Non-Reg.

Sol:

- 1- Assume that L (Palindrome) is Reg.
- 2- An FA for ' L ' a FA consist of states ($N =$)
- 3- Take a string ' w ' for ' L ' of length more than N

Decidability of R.L (Equivalence)

Checking equivalence of R.L.

$$L_{eq} = (L_1 \cap L_2^c) + (L_2 \cap L_1^c)$$

$$= (FA_1 \cap FA_2^c) + (FA_2 \cap FA_1^c)$$

$$\downarrow$$

$$= FA_3$$

Context Free Grammar (CFG)

↳ Generate strings

NV - Non-Terminal $\rightarrow$ variable (capital letters)
e.g. A, B, C, ..., X, Y, Z
 $\hookrightarrow$ Need to be defined further

T 2-Terminal $\rightarrow$ Primitive symbols
 e.g. $a, b, c, \dots, x, y, z$
 $\hookrightarrow$ no need to define further

P 3- Rules Production $A \rightarrow \alpha$
 $\downarrow$
 N.T

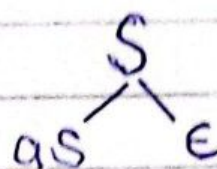
S 4. The starting Symbol (e.g. S)

Date: _____

LHS
(Single Non-Ter.)

RHS
(Ter. + Non Ter.)

$S \xrightarrow{\text{leads to}} aS$ (multiple a)
 $S \rightarrow \epsilon$ (a^*)



$S \rightarrow$ starting state

Derivation Of Strings:

$S \rightarrow aS \rightarrow \epsilon$ $a \cdot \epsilon = a$

OR

$S \rightarrow aS \rightarrow aS \rightarrow aS \rightarrow aS \rightarrow \epsilon$

(capital se end nhi hota)

String with atleast one double letter

$(a+b)^*$ $(aa+bb)$ $(a+b)^*$ R.E

~~$S \rightarrow aaS + bbS$
 $S \rightarrow aS + bS$
 $S \rightarrow \epsilon$~~

$\uparrow$
 $S \rightarrow ADA$
 $A \rightarrow aA$
 $A \rightarrow bA$
 $A \rightarrow \epsilon$
 $D \rightarrow aa$
 $D \rightarrow bb$

Can not derive same production twice

y: _____

Date: _____

$$S \rightarrow X bbb X$$

$$X \rightarrow \Lambda | aX | bX \quad (a+b)^*$$

start & end different

$$S \rightarrow aX.b$$

$$S \rightarrow bX.a$$
~~$$S \rightarrow aX$$~~

$$X \rightarrow \Lambda | aX | bX$$

$$a(a+b)^*b$$

$$b(a+b)^*a$$

$$S \rightarrow A | B$$

$$A \rightarrow aXb$$

$$B \rightarrow bXa$$

$$X \rightarrow \Lambda | aX | bX$$

do not end with ba. $(a+b)^*aa, bb, ab$

~~$$S \rightarrow Xaa | Xbb | Zab$$~~
~~$$X \rightarrow \Lambda | aX | bX$$~~
~~$$Y \rightarrow \Lambda | aX | bX$$~~
~~$$Z \rightarrow$$~~

$$S \rightarrow Xaa | Xbb | Xab$$

$$X \rightarrow \Lambda | aX | bX$$

ay: _____

Date: _____

Which lang. do these CFG's define.

~~$S \rightarrow \epsilon$~~

$S \rightarrow abS$

$S \rightarrow ab \quad (ab)^+$

$a^n b^n$

$S \rightarrow aSb \mid \epsilon \mid ab$

$S \rightarrow aSb$

$S \rightarrow \Lambda$

$S \rightarrow ab$

CFG For Equal

~~$S \rightarrow aAb$~~

~~$S \rightarrow bBa$~~

~~$A \rightarrow$~~

$S \rightarrow aB$

$S \rightarrow bA$

$A \rightarrow a$

$A \rightarrow aS$

$A \rightarrow bAA$

$B \rightarrow b$

$B \rightarrow bS$

$B \rightarrow aBB$

$S \rightarrow SASBS$

$S \rightarrow SBSAS$

$S \rightarrow \epsilon$

$A \rightarrow a$

$B \rightarrow b$

BNF Notation:

Day: _____

Date: _____

Backus $\leftarrow$ ^{Not} ^{Form} **BNF** NOTATION

$S \rightarrow aB \mid bA$

$\langle S \rangle ::= a \langle B \rangle \mid b \langle A \rangle$

$A \rightarrow a \mid aS \mid bAA$

$\langle A \rangle ::= a \mid a \langle S \rangle \mid b \langle A \rangle \langle A \rangle$

CFG For Identifier:

- Identifier $\rightarrow$ ALPHA $\rightarrow$ Alphanumeric
- Alpha $\rightarrow A \mid B \dots \mid Z \mid a \mid b \mid c \dots \mid z$.
- Alphanumeric $\rightarrow$

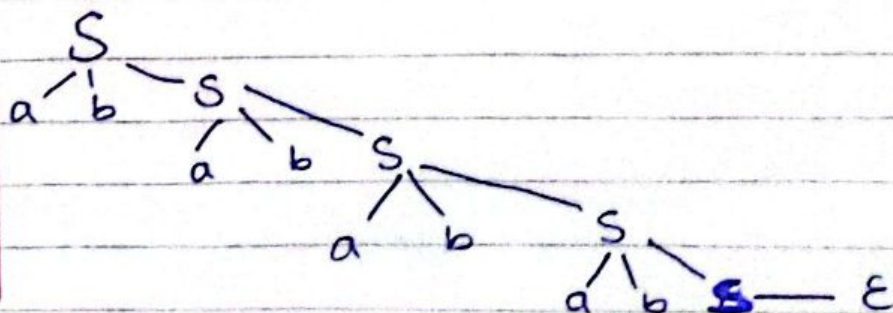
Parse Tree

Syntax tree, generation tree, Prod. Tree, deriv. tree

S
|
abS
|
ε

Example:

- $S \rightarrow abS \mid \Lambda$
- Derivation of abababab



ay: _____

Date: _____

Derivation

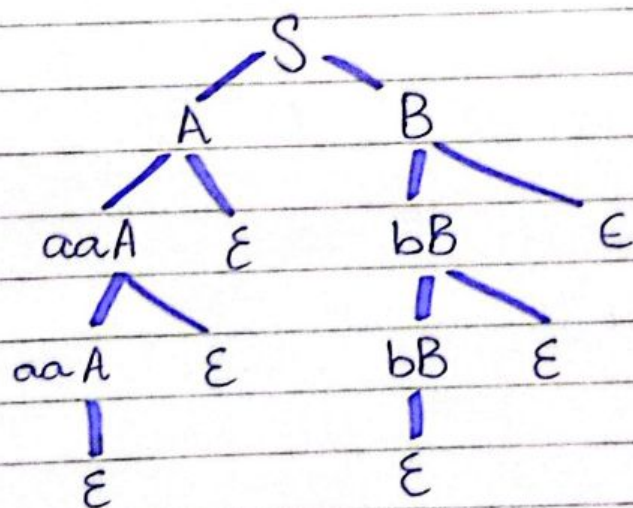
Left-Most D / Right-Most D: (Parsing Techniques)

$$S \rightarrow AB$$

$$A \rightarrow aaA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

'aaaaabb' $\in L$?



String Generator / Derivation through rules.

$L(G)$:-

$$S \rightarrow ABA$$

$$A \rightarrow a|b$$

$$B \rightarrow y|yz$$

$$y \rightarrow \epsilon$$

$$z = b$$

Day: _____

Date: _____

'baab'

LMD:

$S \rightarrow A$	B	A
$\rightarrow alb$	B	A
$\rightarrow alb$	YY/ZZ	A
$\rightarrow alb$	$aalbb$	A
$\rightarrow alb$	$aalbb$	alb
$\rightarrow b$	aa	b

$\rightarrow baab$

RMD:

$S \rightarrow A$	B	A
$\rightarrow A$	B	alb
$\rightarrow A$	yy/zz	alb
$\rightarrow A$	$aalbb$	alb
$\rightarrow alb$	$aalbb$	alb
$\rightarrow b$	aa	b

$\rightarrow baab$

Ambiguous

(1 se zyda parse tree ban ne chahiye)

Day: _____

Date: _____

Example:

$$S \rightarrow bSb$$

$$S \rightarrow Y$$

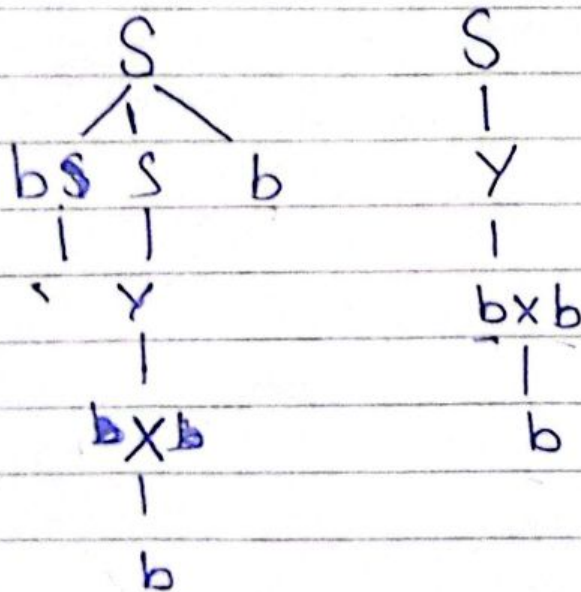
$$Y \rightarrow bXb$$

$$Y \rightarrow x$$

$$X \rightarrow aXa$$

$$X \rightarrow b$$

"bbb"



Total Language Tree:

string bante Sale possible jetna
ha se bana dega.

Day: _____

CH-B

Date: _____

RL ϵ CFG's

Theoram:

$$FA \rightarrow CFG$$

- Non ϵ terminal symbol will get converted in states

$$\begin{array}{l} X \rightarrow \alpha y \\ x \rightarrow b y \\ (x) \xrightarrow{a,b} (y) \end{array}$$

- Recursive

$$(x) \xrightarrow{a} (x) \quad X \rightarrow aX$$

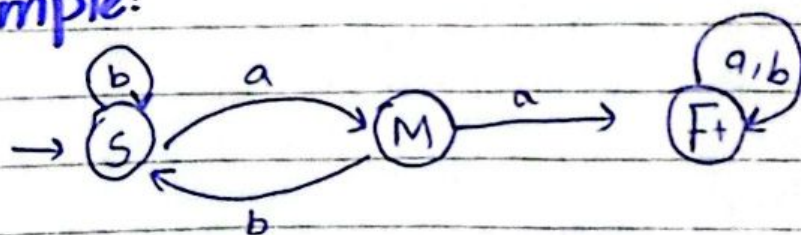
- $NT \rightarrow \epsilon$

Proof:

$$FA \rightarrow CFG$$

$$CFG \rightarrow FA$$

Example:



Day: _____

Date: _____

$$\rightarrow S \rightarrow bS$$

$$S \rightarrow aM$$

$$M \rightarrow bS$$

$$M \rightarrow aF$$

$$F \rightarrow aF | bF | \Lambda$$